Aqualert

Supporting Safety Around Water

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Figure 1: Aqualert Logo

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Exhibit Overview

This project is a wearable water-safety bracelet system designed to help parents monitor children in environments like pools, beaches, and crowded outdoor spaces. The system uses a paired bracelet and app to track heart rate, motion, and water exposure, sending real-time alerts to caregivers if a child enters water unexpectedly or shows signs of distress. The prototype explores how wearable technology can improve safety while remaining simple, intuitive, and accessible during high-stress situations. At this stage, it is simply an idea, but with how technology is evolving, this is a practical idea with a high possibility of it being made.

What it Augments

The bracelet augments parental awareness and child safety by providing additional real-time monitoring in water-related environments. Instead of replacing supervision, it acts as an extra layer of protection that helps caregivers react faster in emergencies. The technology enhances communication between parent and child through wearable alerts and sensor-based monitoring.

Who is it For

This system is designed for parents, guardians, and caregivers responsible for young children around water. It is especially intended for use in places where distractions, crowds, or limited visibility can make supervision more difficult, such as swimming pools, beaches, water parks, and outdoor recreational areas.

How it Works

The system includes a wearable bracelet worn by the child and a connected mobile app used by the parent or caregiver. The bracelet monitors heart rate, movement, and water exposure using built-in sensors, then sends that information to the app in real time. If the system detects unexpected water entry, unusual movement, or signs of distress, the app immediately sends alerts and emergency notifications to the caregiver's phone through vibration, sound, and visual warnings. The app also allows parents to monitor safety metrics live, helping them respond quickly in potentially dangerous situations.

Process & Critical Making

The prototype is currently in an early but functional stage. I designed and built a basic working model of the bracelet that demonstrates the overall concept and key safety features. In addition to the physical prototype, I also created a functional app prototype in Figma that shows how the system would pair with the bracelet and display real-time safety data, alerts, and emergency notifications on a caregiver's phone.

Throughout the process, I faced challenges balancing simplicity with functionality. I had to rethink certain features to avoid making the bracelet bulky or overwhelming while keeping it useful and effective. This iterative process also extended into the app design, where I focused on making the interface clear and easy to interpret in high-stress situations.

Accessibility, Usability, & User Experience

A major focus of this project was creating a system that feels intuitive and easy to use during stressful situations. Since emergencies require quick reactions, the bracelet is designed with simple alerts and minimal interaction needed from the user. Accessibility also influenced decisions about comfort, readability, and ease of understanding for both children and adults. The project reinforced how good user experience design is not only about adding features, but about making those features clear, reliable, and usable in real-world situations.

Why it Matters

Drowning and water-related accidents can happen quickly and unexpectedly, especially with young children. This project explores how wearable technology can provide additional support for caregivers and potentially improve response time during emergencies. At the same time, it raises important questions about privacy, biometric data collection, and dependence on technology. The bracelet is meant to support active supervision rather than replace it, encouraging safer environments while demonstrating how technology can be thoughtfully integrated into everyday life.

Prototype Media

The Bracelet

The bracelet is a wearable safety device designed for children that tracks heart rate, movement, and water exposure to help monitor potential risk in real time. In theory, it would pair with a caregiver's phone app to send alerts and share live safety data when needed. The design is intended to be lightweight, durable, and comfortable for everyday wear in water-related environments like pools and beaches. It would also be customizable by color, allowing children to choose a style that feels more personal and less clinical while still serving a critical safety function.

This is an example of the standard black model:

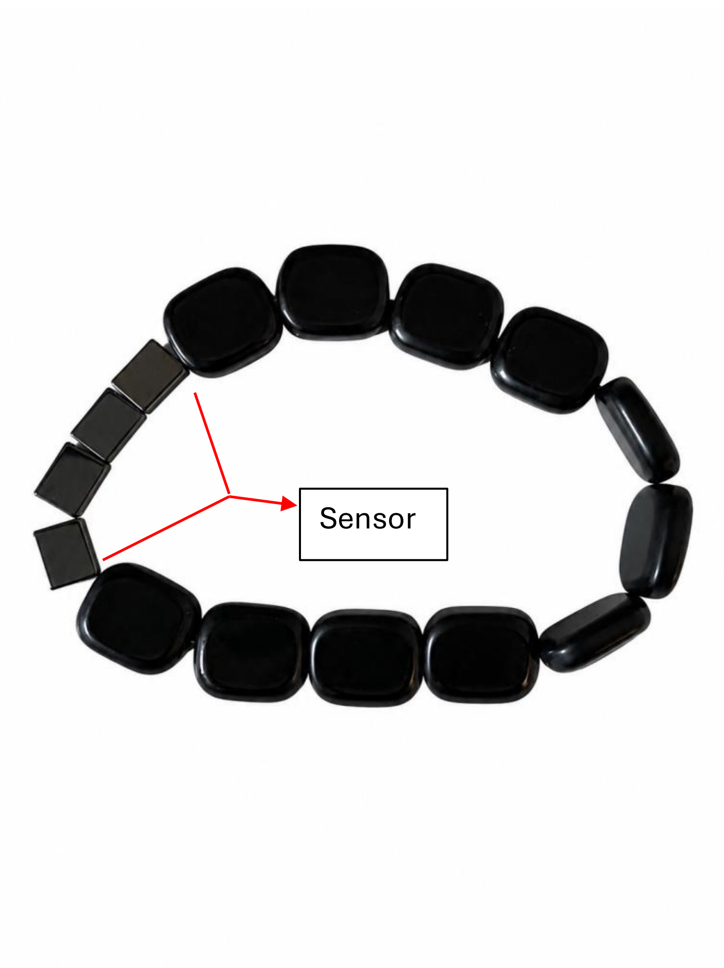


Figure 2: Aqualert bracelet prototype

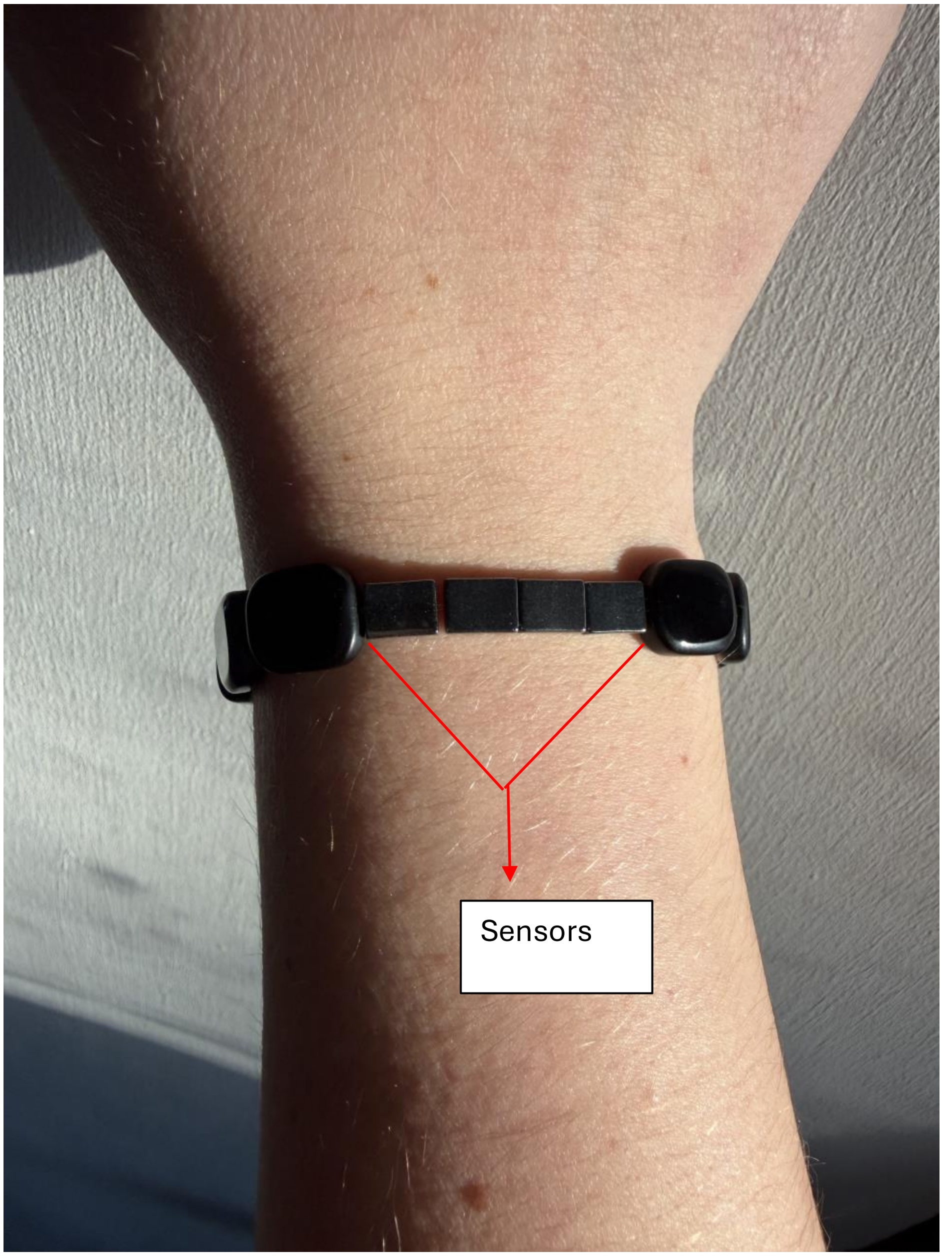


Figure 3: Aqualert bracelet prototype on wrist

The App

I designed a companion mobile app in Figma that pairs with the bracelet and shows how caregivers would interact with the system in real time. The app translates data like heart rate, movement, and water exposure into a simple interface on the parent's phone, along with live monitoring and emergency alerts. It is designed to be clear and easy to read so caregivers can respond quickly in high-stress situations.

Below are screen captures from the app, but if you would like to see how the app functions in real time, please proceed to this link: [Aqualert Bracelet App]

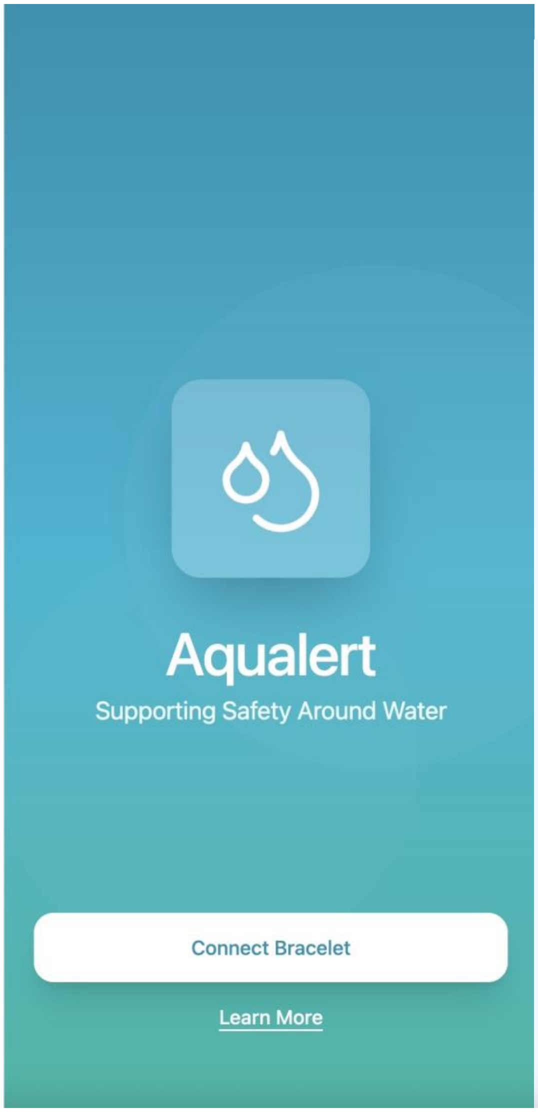


Figure 4: Aqualert app "Welcome Screen"

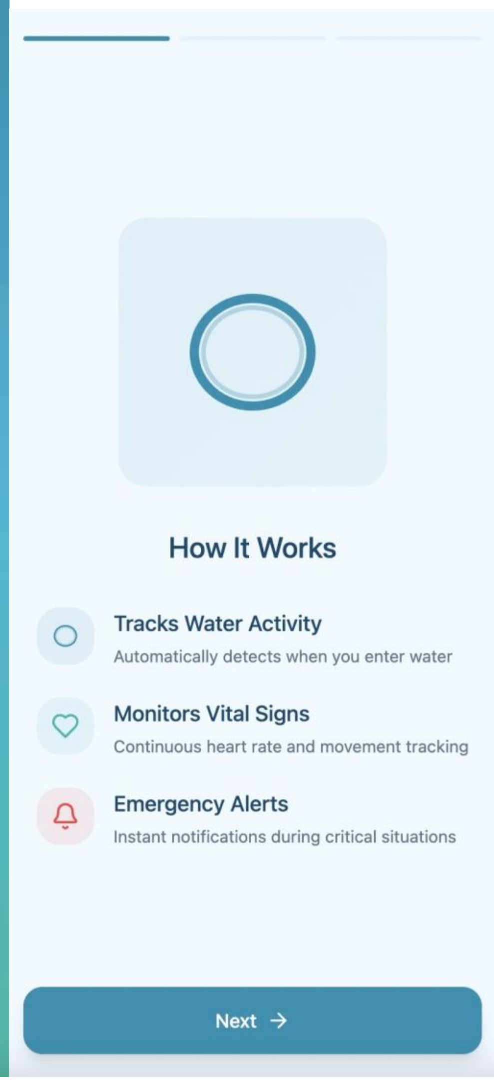
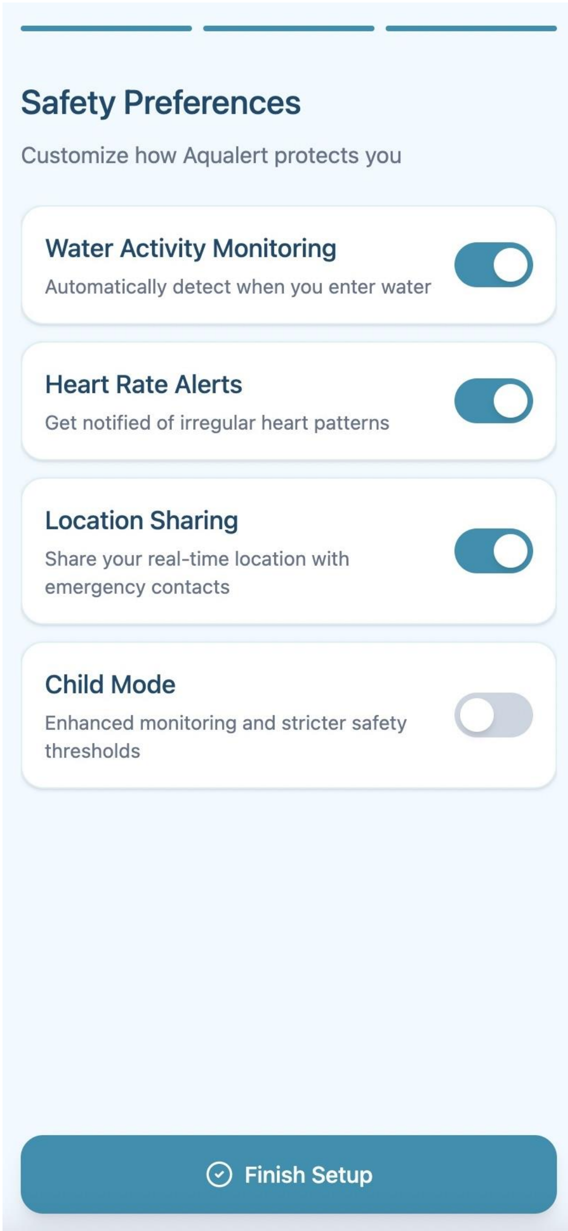


Figure 5: Aqualert app "How it Works"



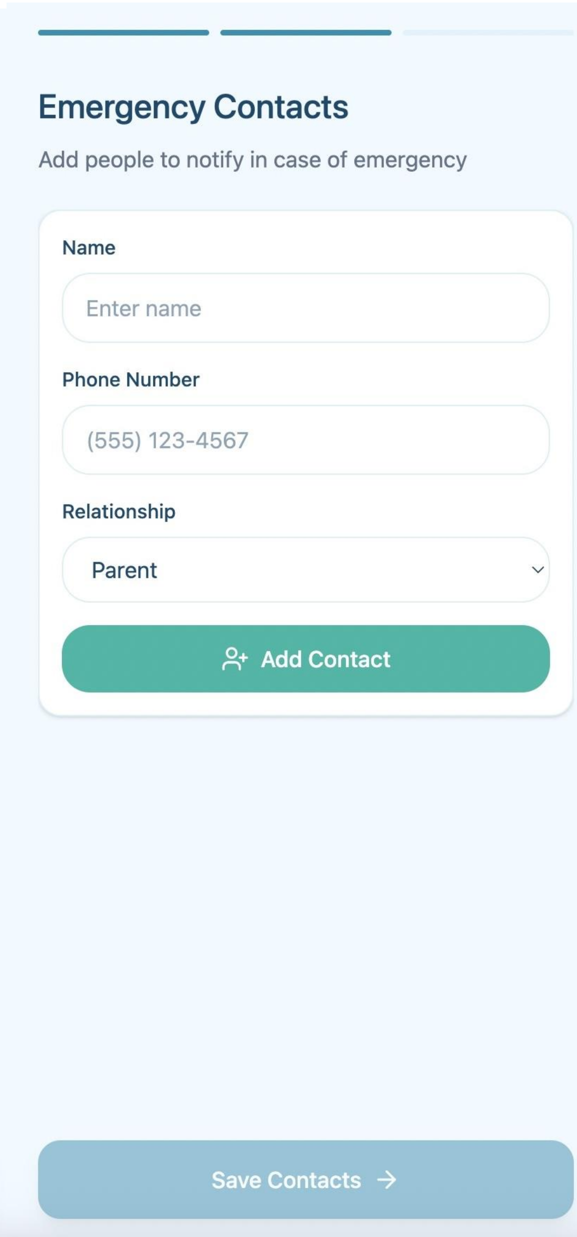


Figure 6: Aqualert app "Safety Preferences"

Figure 7: Aqualert app "Emergency Contact"

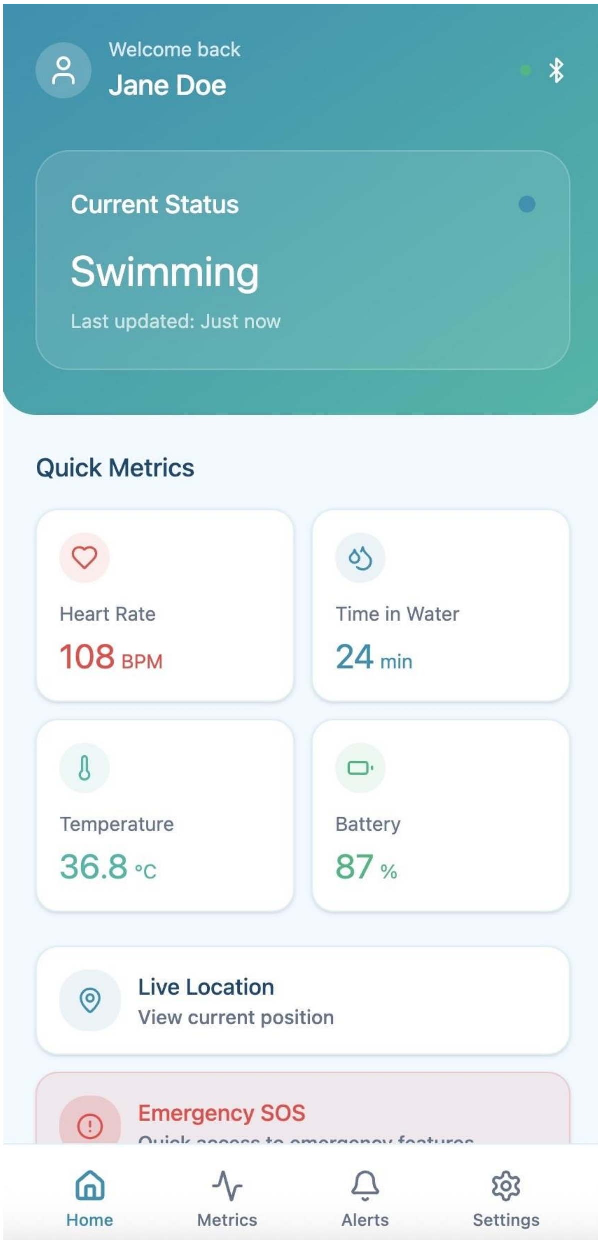


Figure 8: Aqualert app "Home Screen"

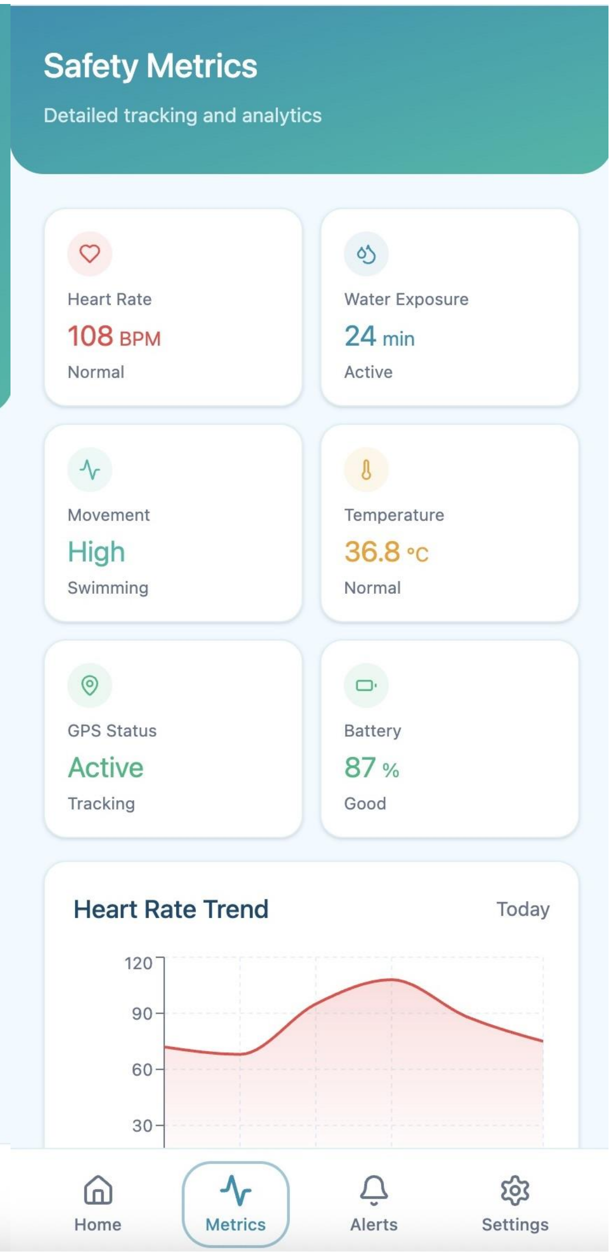


Figure 9: Aqualert app "Safety Metrics"

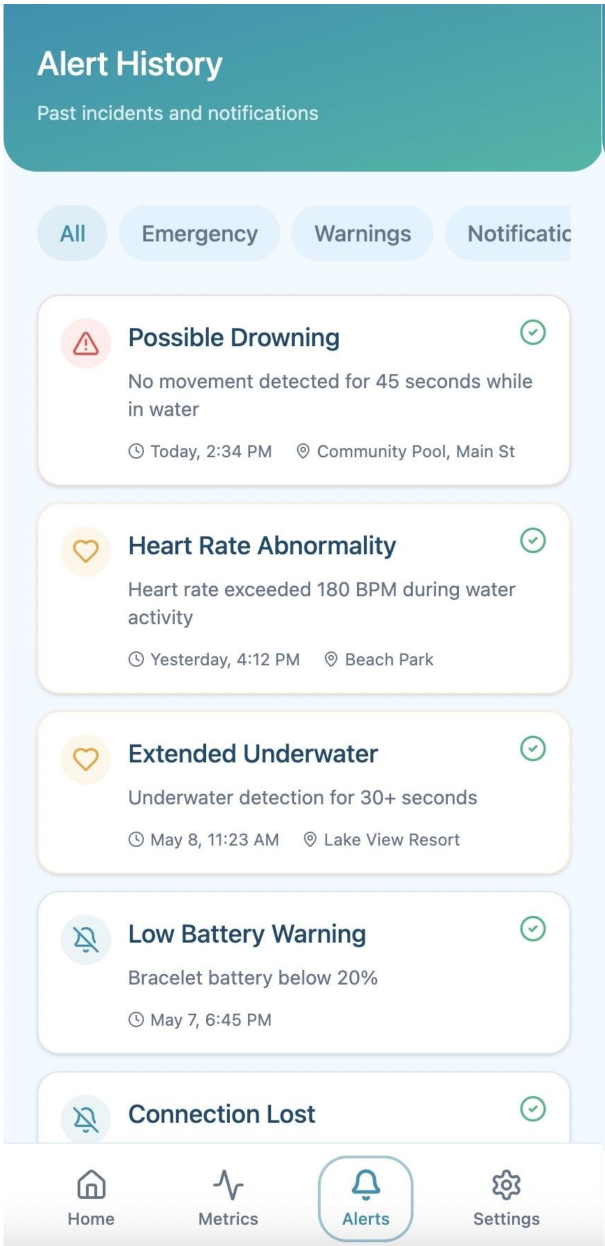


Figure 10: Aqualert app "Alert History"

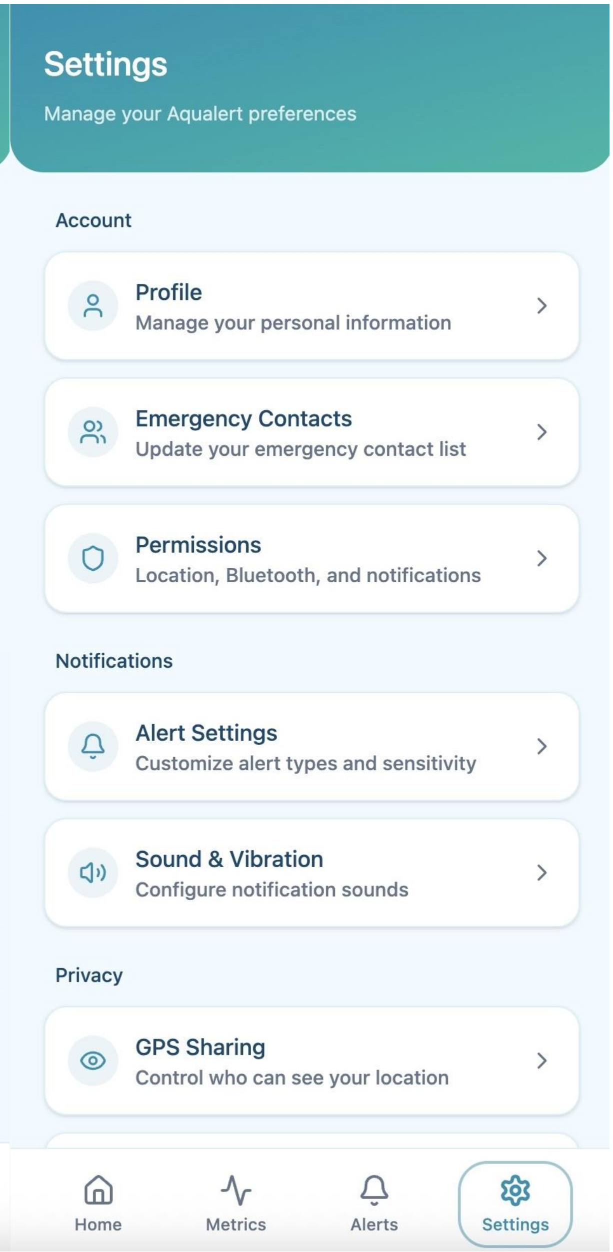


Figure 11: Aqualert app "Settings"

Reflection

My solution, Aqualert, is in response to the broader sociotechnical system of water safety. Safety is determined through a combination of human monitoring, environmental factors, and available safety tools. The overarching problem being solved here is the commonality of preventable child drownings in systems such as pools and open water that requires the utmost vigilance and can be compromised in moments. Having worked as a lifeguard for nearly six years, I have seen firsthand how easily and quickly such situations can unfold, even when adults think they are watching attentively. The system is designed to augment care given by children and their parents/guardians. A primary issue the system addresses is the gap between the idea of "being watched" and having attentive presence at every moment; when crowded, distracted, or in a different location, the delay time for noticing trouble and intervening drastically increases.

The design is fundamentally rooted in accessibility and UX, and has been shaped to be functional during a high-stress, time-sensitive emergency. This entails making the device extremely easy to interpret in both normal situations where the functionality needs to become invisible and in moments of panic and high emotion where an immediate indication for intervention should stand out clearly. Accessibility contributes to that the system does not require the user to know how to use a complex interface, and is easily understood by a variety of users. UX is key as the app is reliant on rapid interpretation of data and minimum distraction during emergencies; although AI is not a core part of the current prototype, the system can potentially become more effective with integration of smart detection algorithms that would ideally identify true danger from non-emergencies. Ultimately, all of these competencies contribute to the overarching goal of producing an alert system for the pool that minimizes detection to intervention time through an understanding of the user experience.

I have progressed toward the solution through the process of iterative simplification of the initial idea. Originally, I was contemplating a more elaborate wearable using a rubber-band system that held the components. As the physical prototype and the associated Figma app prototype developed, I found the need for minimalism and simplicity became apparent. Working with the bracelet forced me to think about real-world issues such as how it feels on the arm and wearability, thereby limiting the scope of the design. Simultaneously, the app development led to understanding how crucial an uncluttered interface is during emergency situations. The result over time is a slimmed-down system for both the bracelet and app, leaving only the most necessary information and functions. This process allowed me to appreciate how prototype development also involves subtraction, paring away what isn't essential until a function is truly clear and concise.

This project feels very personal to me due to my involvement in lifeguarding and the continuous experience I have had with the world of water safety. Living and working in Arizona before college, where homes almost invariably include a pool, I have witnessed an inordinate number of unsupervised kids around pools and a subsequent lack of response that immediately escalates into emergencies. This was the driving force behind the design; this was a real problem to me that I felt was solvable through an effective augmented design system. The development process allowed me to understand the practical application of both accessibility and UX to a critical real-world system, in turn enabling me to develop skills in prototyping, systems thinking and problem translation into both a physical and digital design. I look forward to continuing iterating the design of the wearable as well as more in-depth interface development, and potential incorporation of more advanced alert systems. The design is a fundamental lesson to me on the direct relation between design, critical response, and milliseconds of human attention.